UBFC-RPPG Dataset

We introduce here a new database called **UBFC-rPPG** (stands for Univ. Bourgogne Franche-Comté Remote PhotoPlethysmoGraphy) comprising of two datasets which are focused specifically for rPPG analysis.

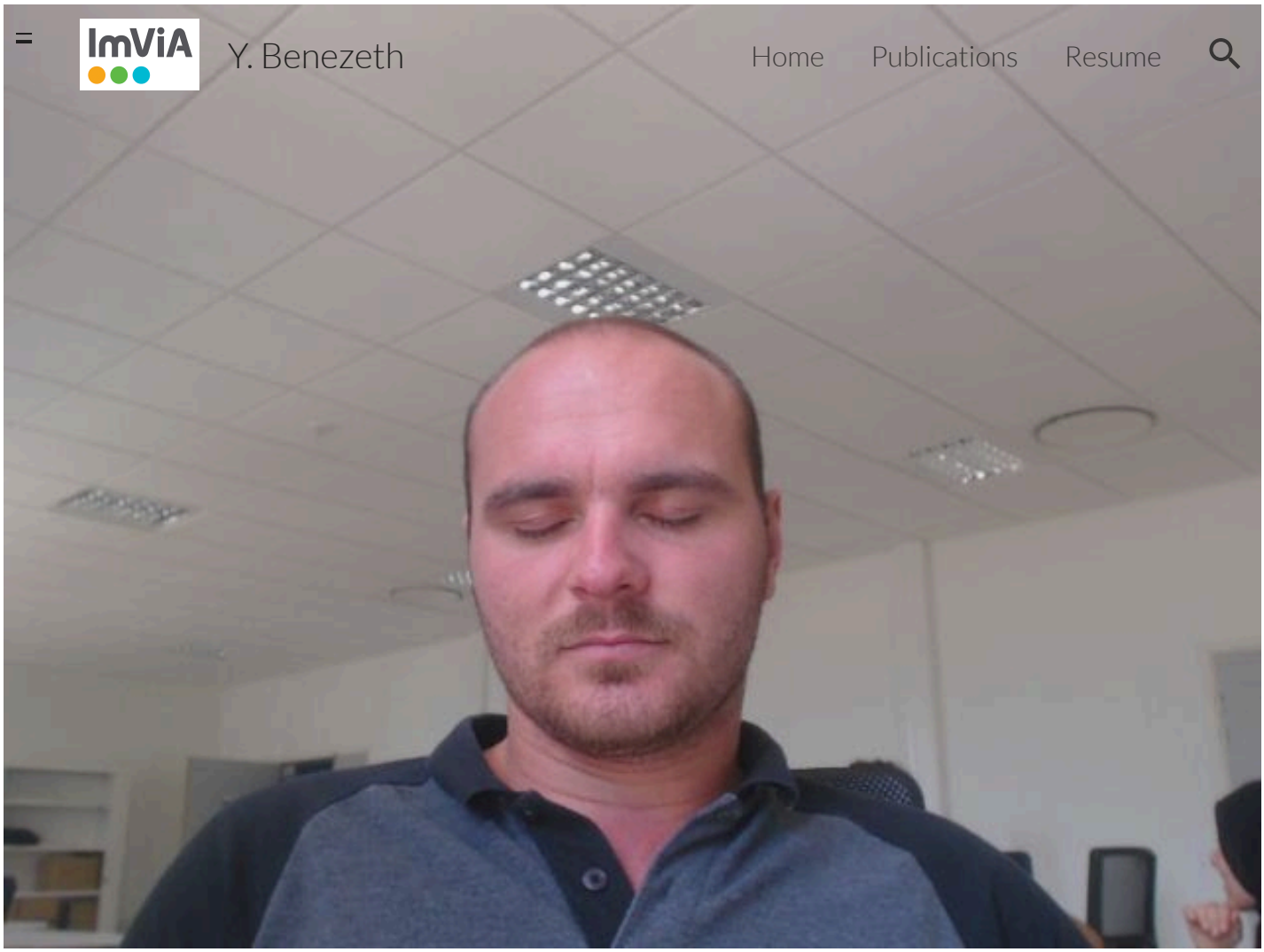
The UBFC-RPPG database was created using a custom C++ application for video acquisition with a simple low cost webcam (Logitech C920 HD Pro) at 30fps with a resolution of 640x480 in uncompressed 8-bit RGB format. A CMS50E transmissive pulse oximeter was used to obtain the ground truth PPG data comprising of the PPG waveform as well as the PPG heart rates. During the recording, the subject sits in front of the camera (about 1m away from the camera) with his/her face visible. All experiments are conducted indoors with a varying amount of sunlight and indoor illumination.

Link to download the complete video dataset is available below. A basic Matlab implementation is also be provided to read ground truth data acquired with a pulse oximeter.

You might also find **UBFC-Phys** interesting, which can also be used for rPPG algorithm research (more info here). Alternatively, consider **IMVIA-NIR**, a publicly available dataset designed for NIR rPPG research (more info here).

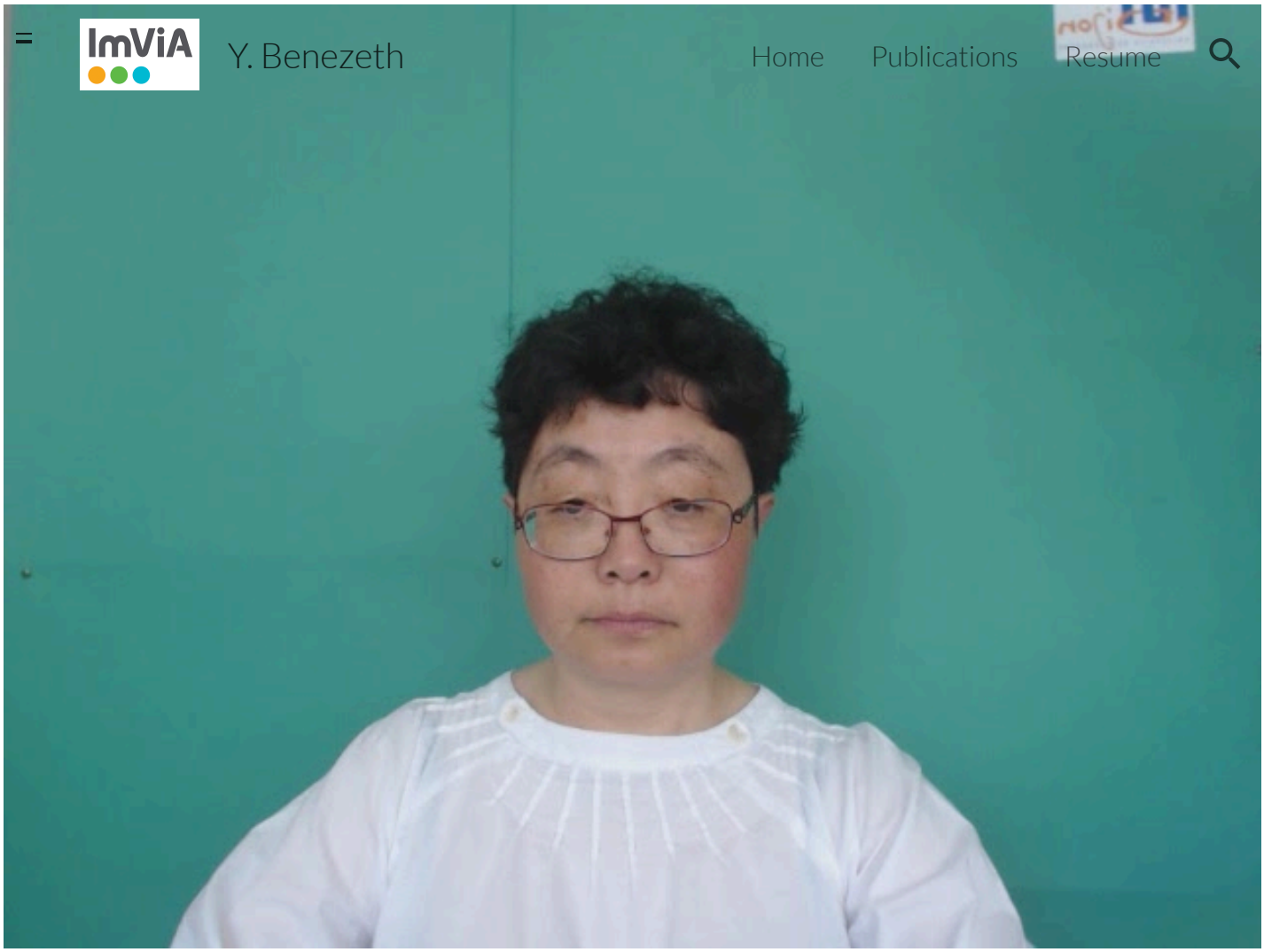
Dataset 1 (simple)

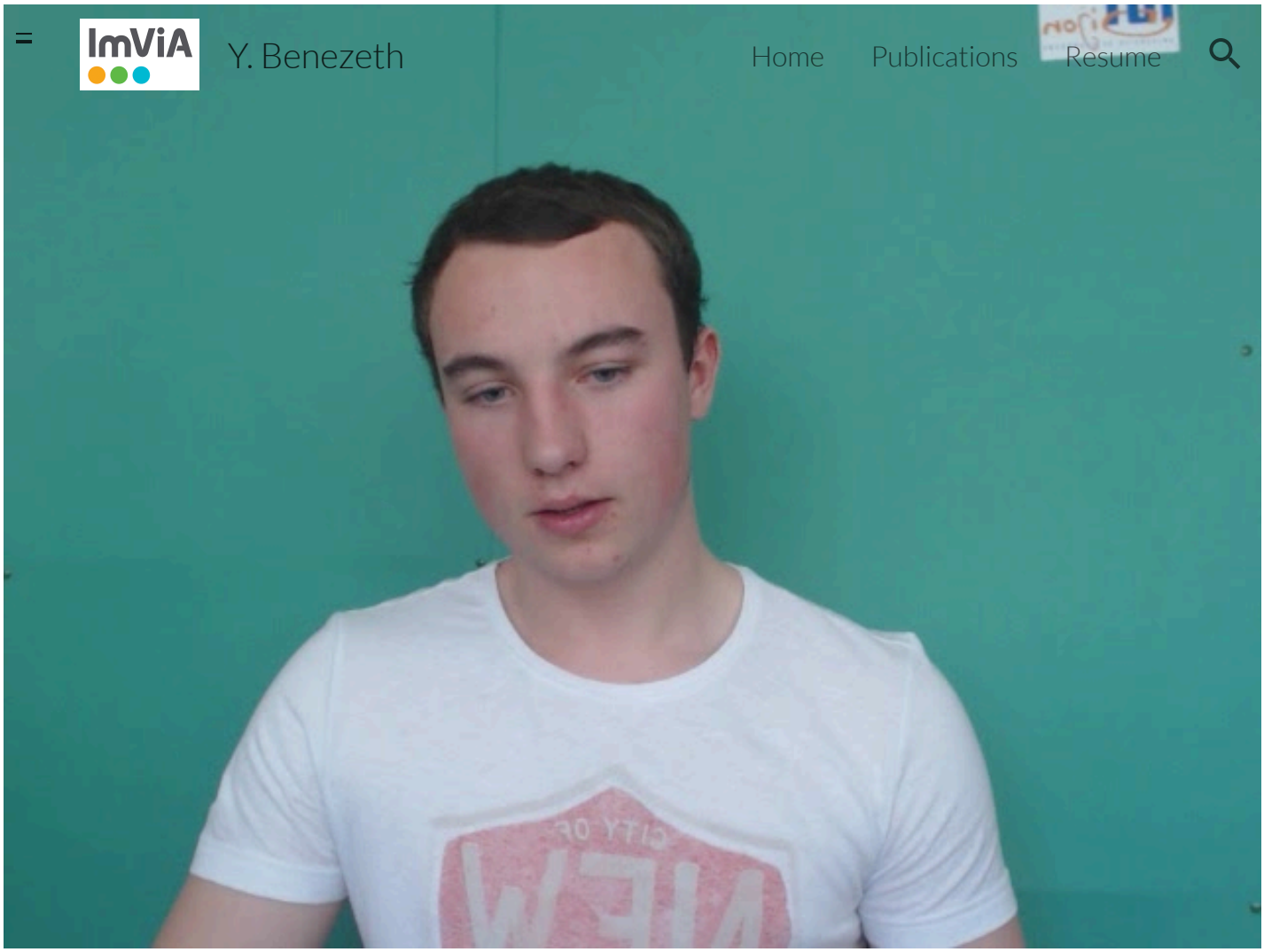
In this first dataset, participants were asked to sit still but some videos present significant movement (especially at the beginning of the sequence). The dataset is composed of 8 videos (about 16500 frames).



Dataset 2 (realistic)

For the second dataset, the subject sits in front of the camera (about 1m away from the camera) with his/her face visible and is required to play a time sensitive mathematical game that aimed at augmenting their heart rate while simultaneously emulating a normal human-computer interaction scenario. 42 videos (among 46 videos) of this dataset are shared for research purpose.





Contact and download

Data have been recorded by R. Macwan and Y. Benezeth. Link to download the video dataset and a basic Matlab and Python code to read ground truth data can be found here.

Please contact if needed:

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Reference

If you use this dataset, please cite this paper:


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S. Bobbia, R. Macwan, Y. Benezeth, A. Mansouri, J. Dubois, "Unsupervised skin tissue segmentation for remote photoplethysmography", Pattern Recognition Letters, 2017.
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